

Alan Braz

Research Software Engineer / AI Systems Architect / Generative AI Leader

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+ Ask anything



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SUMMARY

Senior Software Engineer and AI Engineering Manager with **20+ years** designing and leading enterprise-grade AI, Cloud, and Software Engineering solutions. I specialize in transforming complex research into scalable production systems, building architectures for highly regulated industries, and leading multidisciplinary engineering teams. I bring deep expertise in AI Systems (LLMs, RAG, Guardrails), Hybrid Cloud, and large-scale data platforms with a track record of delivering mission-critical solutions for global clients. **Master's** in Computer Science, **15 patents**, **18 publications**, and extensive leadership across IBM Research initiatives.

CORE EXPERTISE

- AI & GenAI Architecture — RAG pipelines, LLM validation, governance, AI guardrails.
- Enterprise Cloud Architecture — Kubernetes, OpenShift, Multi-Cloud, Terraform.
- Engineering Leadership — Tech strategy, high-performance teams, agile delivery.
- Research to Product — Converting innovation into operational platforms.

PROFESSIONAL EXPERIENCE

IBM Research — Brazil & USA

Research Engineering Manager — AI & Hybrid Cloud

2025 – Present

- Lead engineering for **LLM systems** including RAG pipelines, guardrails, inference optimization, and enterprise-grade validation frameworks.
- Drive adoption of **Generative AI** to increase engineering throughput across all software delivery phases.

Senior Research Software Engineer & Global Tech Lead

2019 – 2025

- **OSDU Cloud Lead (IBM side):** Architected and delivered distributed APIs using **Java/Spring Boot** on Kubernetes/OpenShift, integrating with **Microsoft, AWS, Google** cloud providers. Resulted in a long-term strategic partnership with Schlumberger and the IBM Open Data for Industries product.
- Designed and automated multi-cloud infrastructure using **Terraform**, AWS, and IBM Cloud for Accelerated Discovery Platform.
- Technical lead for converting the research project **Seismic Interpretation Advisor (SIA)** into the commercial **GeoScience Advisor (GSA)**—built as a scalable cloud-native AI system.
- Created and led weekly **global engineering dev-hours**, fostering cross-team architecture + code review culture.

Software Engineering Researcher — IBM TJ Watson (NY HQ)

2017 – 2019

- Built cloud-native front-end and API components using Angular, React, WebComponents.
- Developed ML lifecycle tracking features later incorporated into **IBM Watson Studio**.
- Co-authored and open-sourced **AIMMX**, an AI model metadata extractor published at MSR.

Software Engineering Researcher — Social Data Analytics

2013 – 2016

Architected real-time Big Data pipelines for social analytics during the **FIFA World Cup**, powering live TV broadcast apps; received **two Research Division Awards**.

IT Specialist & Java Architect

2005 – 2013

- JavaEE systems architecture, Agile coaching, ScrumMaster, DevOps and engineering tooling.

PROJECTS & LEADERSHIP

- **TechNights Co-Organizer (2023–2025):** Led 21+ community engineering events with 1200+ participants; topics included AI, Automation, Security, Quantum, Hybrid Cloud.
- **AI Committee Leader (FDC):** Delivered AI strategy sessions to CXO audiences; organized immersions at IBM Chicago, MIT Watson AI Lab, and IBM NY HQ.
- **TLC-BR Co-Chair:** Guided IBM's 3,000-member technical community to develop leadership and engineering excellence.

SKILLS

AI Architecture • GenAI Engineering • LLM Ops • Cloud (AWS, IBM Cloud, OpenShift) • Enterprise Integration • Solution Architecture • DevOps • MLOps • Data Platforms • Compliance-Ready Architectures • Agile Leadership

PATENTS (Selected, 15 Total)

- Hierarchical Dynamic Deployment of AI Model
- AI model metadata extraction systems (AIMMX)
- Conversational AI evaluation frameworks
- Blockchain-based energy optimization
- Social network-based data recovery

PUBLICATIONS (Selected, 18+)

- “AIMMX: Artificial Intelligence Model Metadata Extractor” – MSR Conference
- “Hybrid Architecture for Multi-Party Conversational Systems” – arXiv
- “AI + Neurologist EEG Annotation Evaluation” – EBioMedicine
- “IBM Design Thinking Software Development Framework” – Brazilian Workshop on Agile Methods

AWARDS

- 2x Research Division Award (RDA) – IBM Research
- IBM Open Innovation Community — All-Star
- Best X-Industry App — Watson + GBS Cognitive Challenge

EDUCATION

- Master's in Computer Science (Software Engineering) — UNICAMP
- Bachelor's in Computer Science — UNICAMP

CERTIFICATIONS (Selected)

- GenAI for Business Leaders (Coursera)
- License to Lead – IBM (Leadership Certification)
- IBM Certified Advanced Architect – Cloud v1
- IBM Certified Technical Advocate – Cloud v4
- InstructLab: Democratizing AI Models at Scale (GenAI)
- Sun Certified Enterprise Architect for the Java Platform, Enterprise Edition 5
- Certified ScrumMaster (CSM) - Scrum Alliance

LANGUAGES

Portuguese (Native), English (Fluent)